PARTICLE SWARM OPTIMIZATION (PSO)





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1. PSO is based on the movement and intelligence of swarms, and it applies the concept of social interaction to problem solving.

2. It was developed in 1995 by James Kennedy (social-psychologist) and Russell Eberhart (electrical engineer). *[Kennedy, J. and Eberhart, R. (1995). "Particle Swarm Optimization"*, Proceedings of the 1995 IEEE International Conference on Neural Networks, *pp. 1942-1948, IEEE Press.*

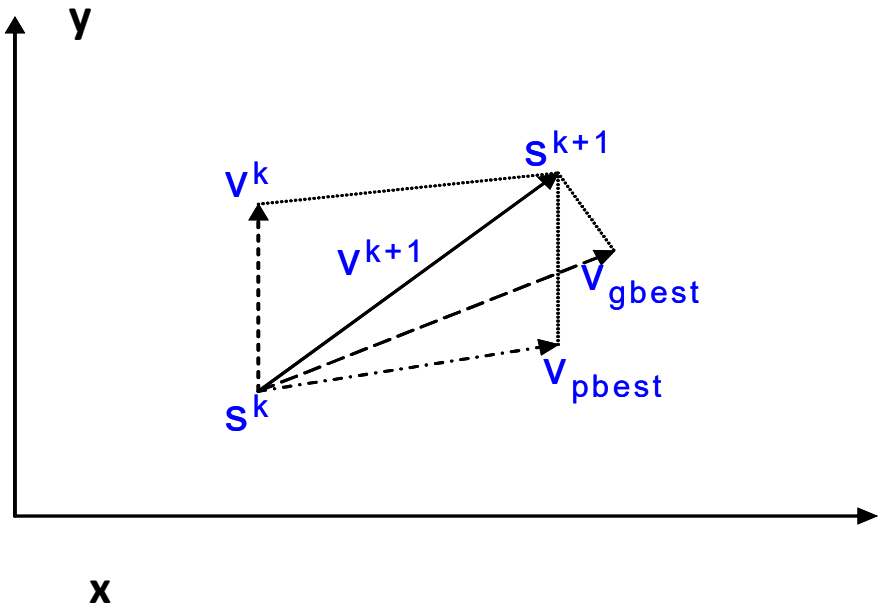
PARTICLE SWARM OPTIMIZATION (PSO)

- In PSO, a swarm of *n* individuals communicate either directly or indirectly with one another in each search directions (gradients).
- It is a population-based search algoritm based on the simulation of the social behavior of birds within a flock
- In PSO, individuals, referred to as particles, are flown through hyperdimensional search space
- The operation of the PSO is based on the neighborhood principle as social network structure.

PSO PROCEDURE

1. Each particle keeps track of its coordinates in the solution space which are associated with the best solution (fitness) that has achieved so far by that particle. This value is called personal best , ***pbest***.
2. Another best value that is tracked by the PSO is the best value obtained so far by any particle in the neighborhood of that particle. This value is called ***gbest***.
3. The basic concept of PSO lies in accelerating each particle toward its pbest and the gbest locations, with a random weighted accelaration at each time step.

Particle Swarm Optimization (PSO)



s^k : current position.
s^{k+1}: modified position.
v^k: current velocity.
v^{k+1}: modified velocity.
v_{pbest} : velocity based on pbest.
v_{gbest} : velocity based on gbest

Each particle tries to modify its position using the following information:

1. the current positions,
2. the current velocities,
3. the distance between the current position and pbest,
4. the distance between the current position and the gbest.

The modification of the **particle's position** can be mathematically modeled according the following equation :

$$V_i^{k+1} = wV_i^k + c_1 rand_1(...)*\left(pbest_i - x_i^k\right) + c_2 rand_2(...)*\left(gbest - x_i^k\right)$$

where,

v_i^k : velocity of particle i at iteration k,
w: weighting function,
c_j : weighting factor,
rand : uniformly distributed random number between 0 and 1,

s_i^k : current position of particle i at iteration k,
$pbest_i$: pbest of partile i,
gbest: gbest of the group.

$$V_{id} = \omega V_{id} + C1rand( )(P_{id}-X_{id}) + C2Rand( )(P_{gd}-X_{id})$$

The significance the above equation can be partitioned into three:

- **1st part** of the equation provides the "flying particles" with a degree of memory capability allowing the exploration of new search space areas.
- **2nd part** is the "cognition" part, which represents the private thinking of the particle itself.
- **3rd part** is the "social" part, which represents the collaboration among the particles.

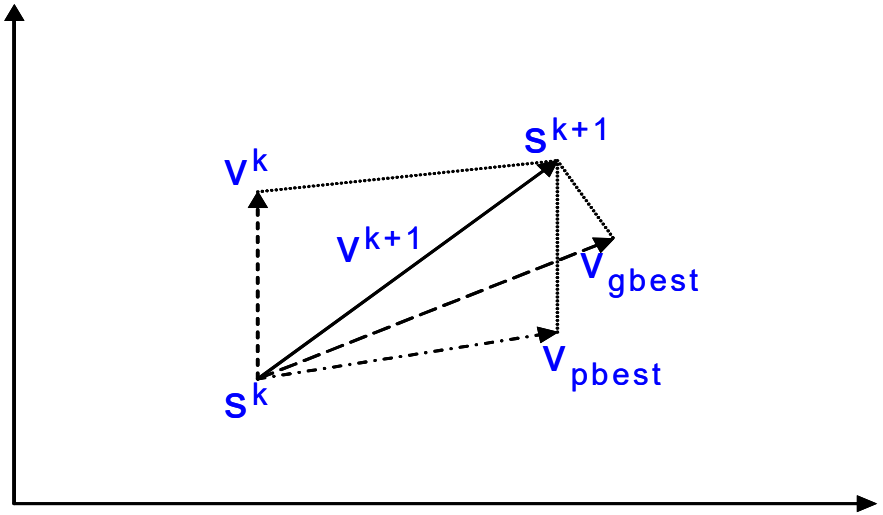
Updating Particle Position

$$S_i^{k+1} = S_i^k + \mathbf{V_i^{k+1}}$$

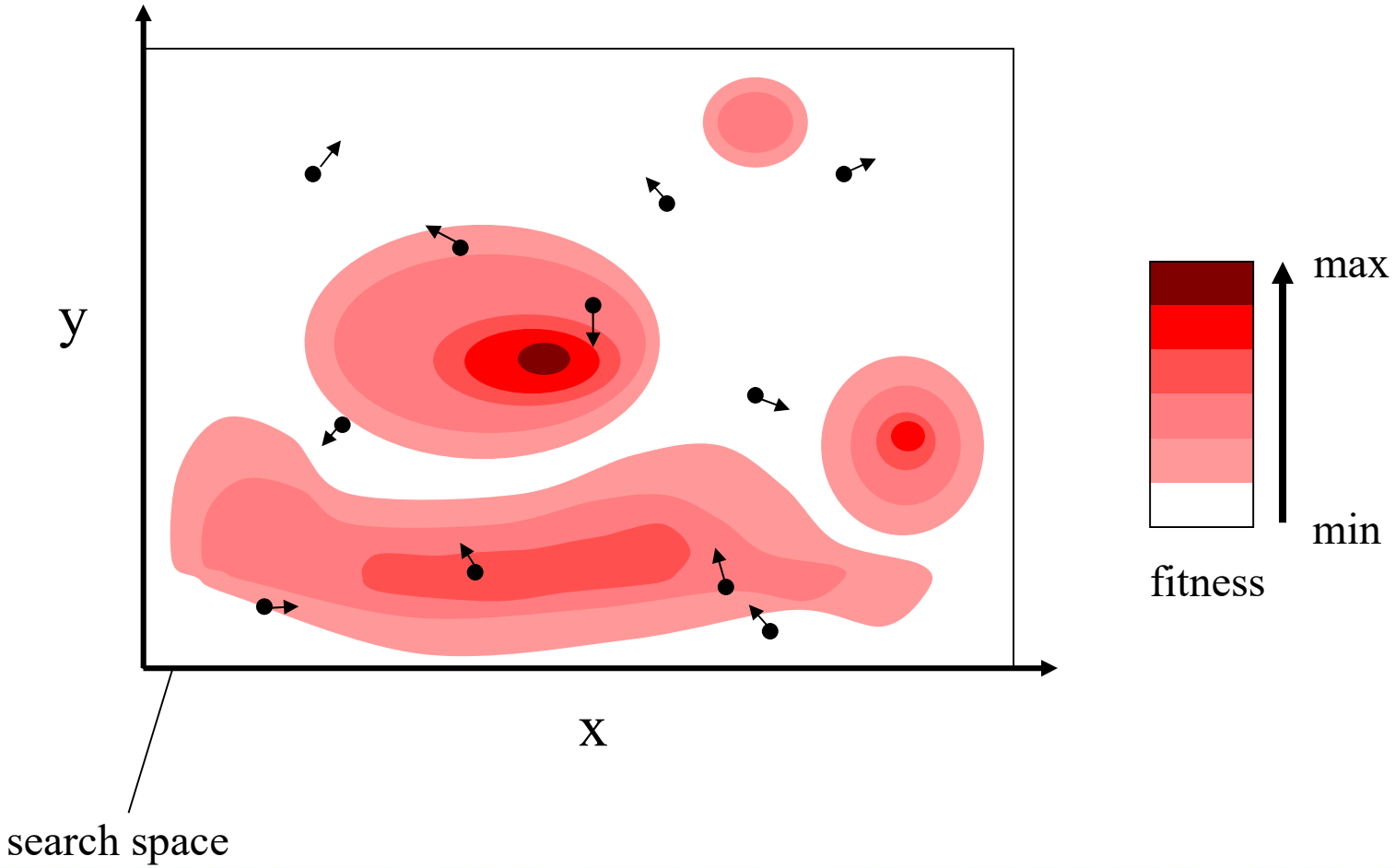
New position

Old position

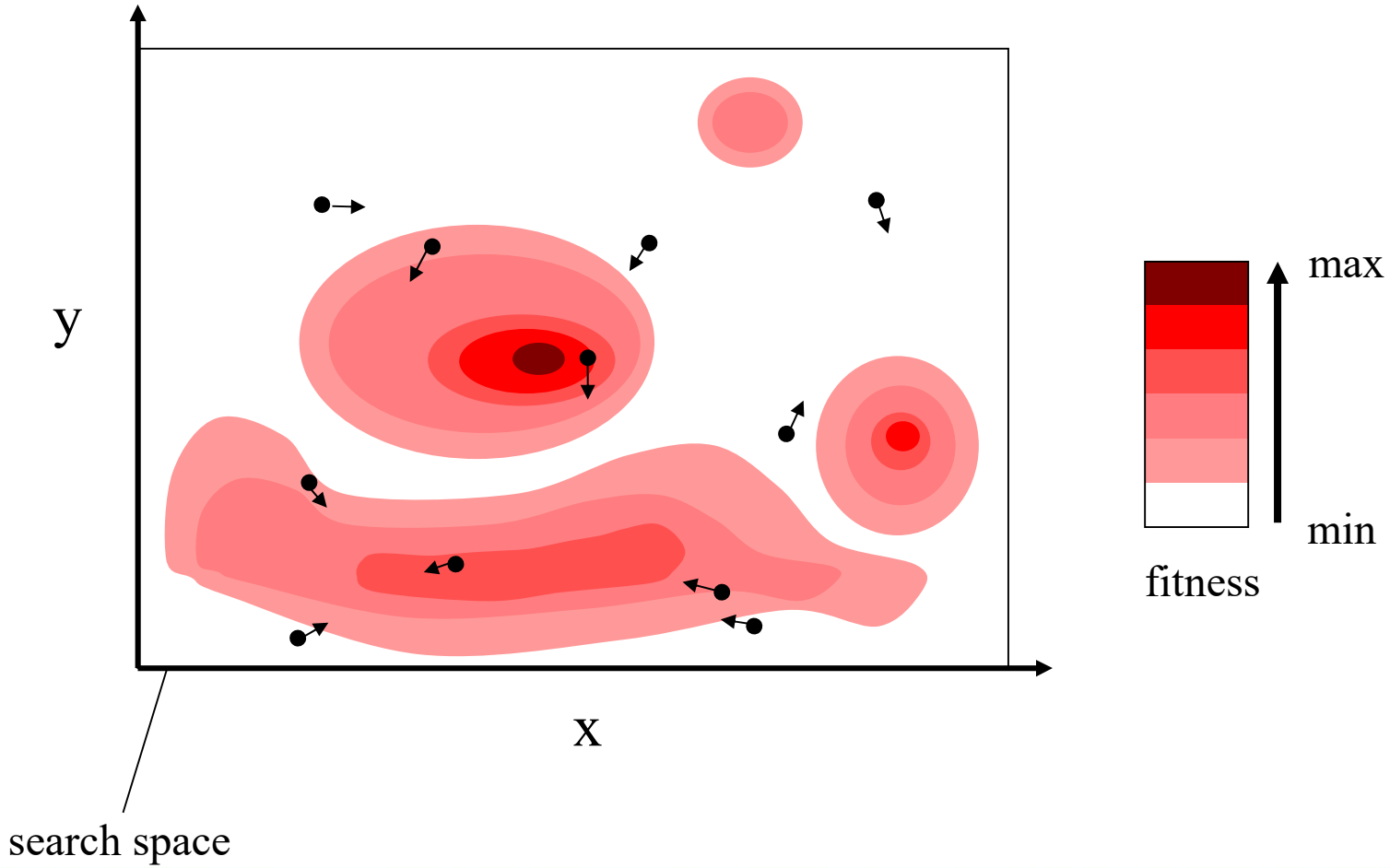
Newly calculated velocity



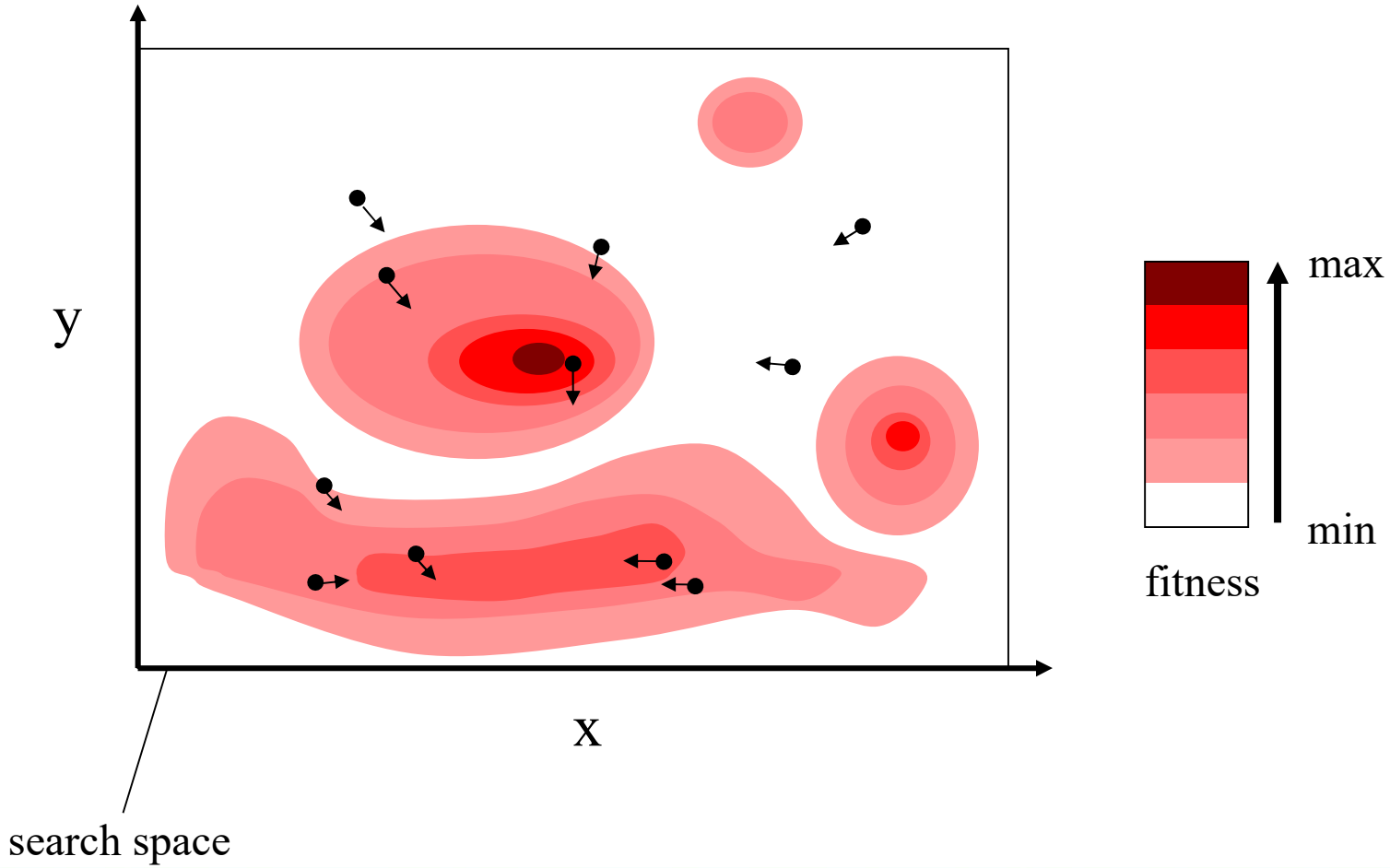
PSO simulation



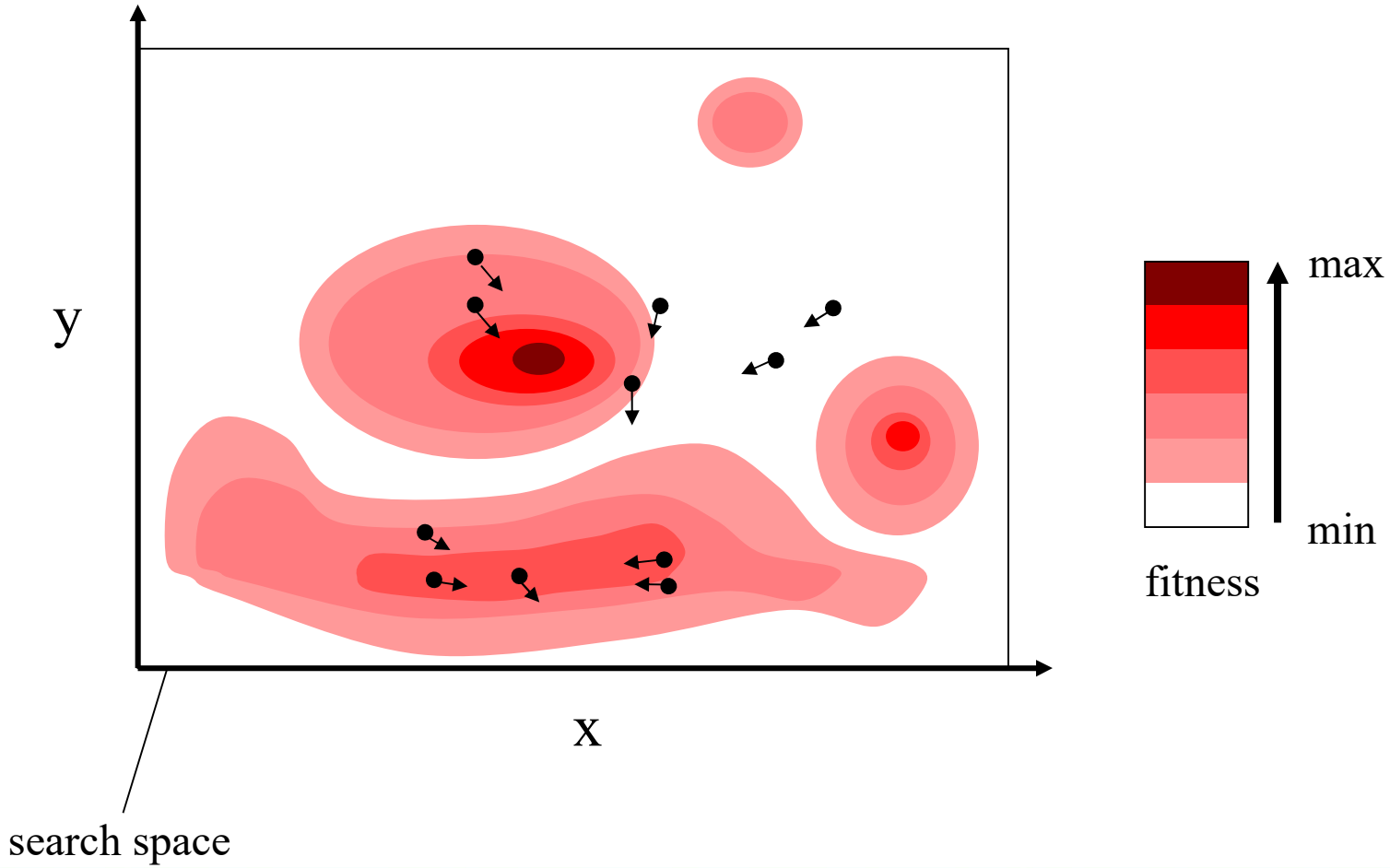
PSO simulation



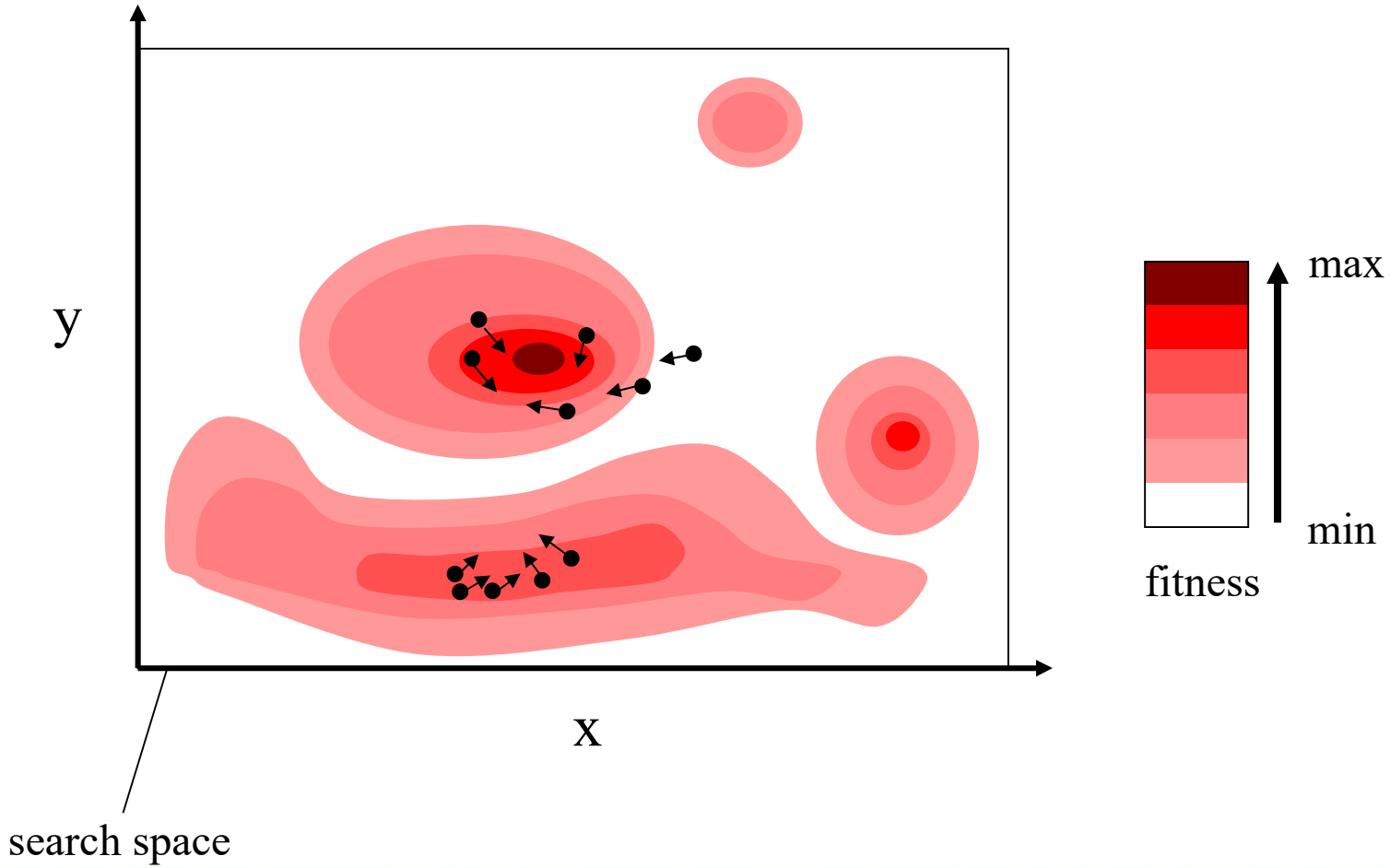
PSO simulation



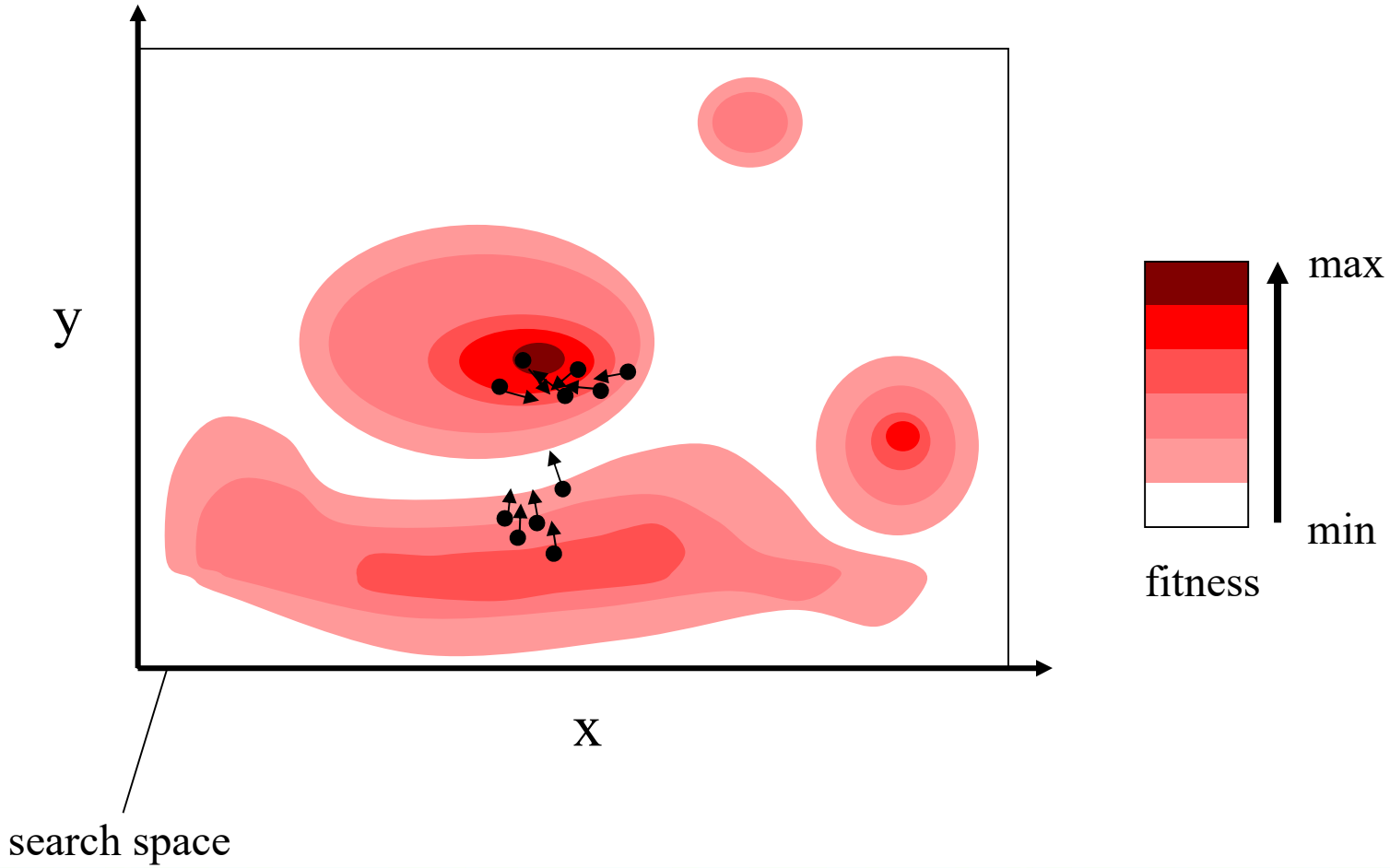
PSO simulation



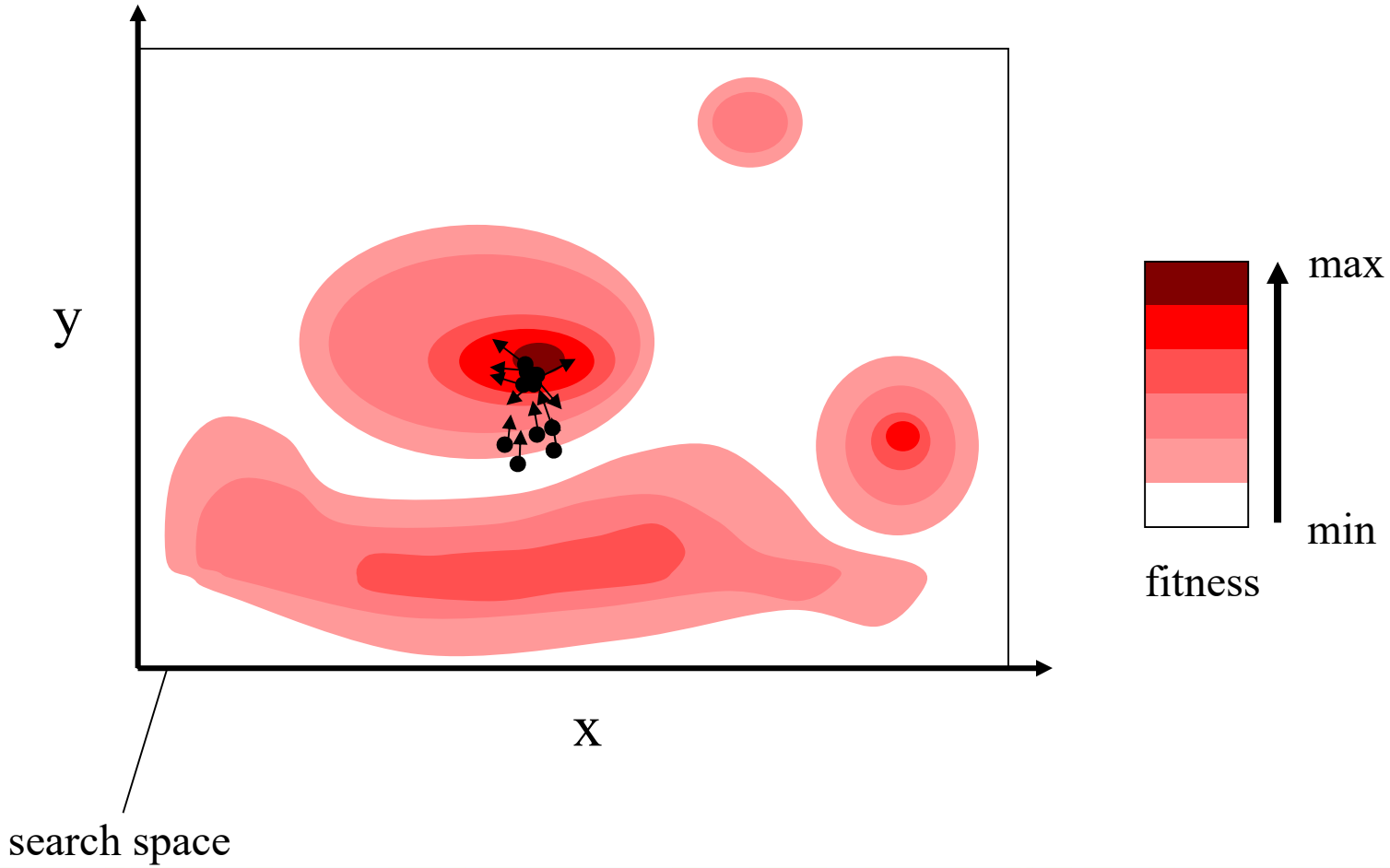
PSO simulation



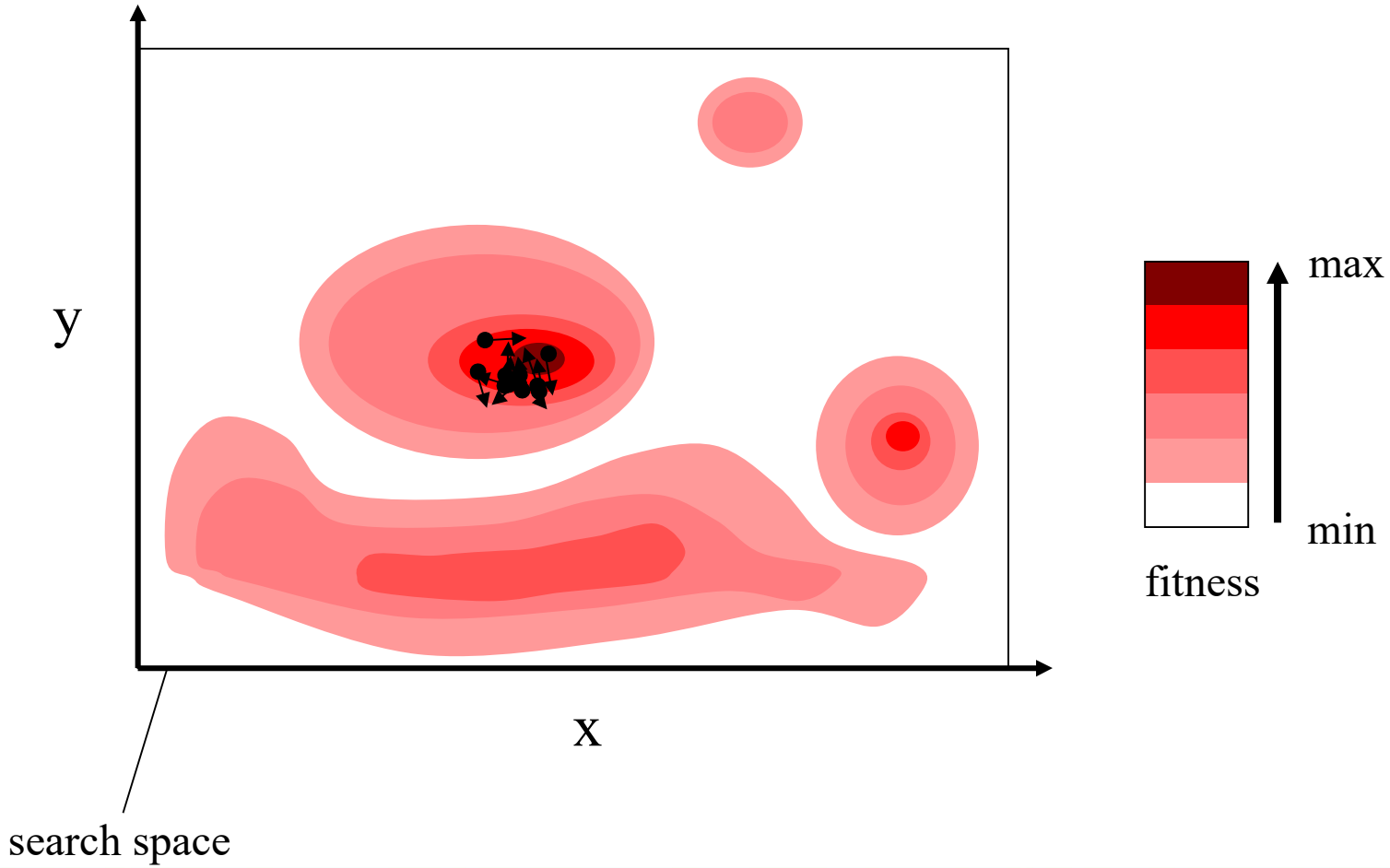
PSO simulation



PSO simulation



PSO simulation



Example Calculation

1. Initialize swarm and set constant variables:

$$c_1 = 1.0$$

$$c_2 = 0.5$$

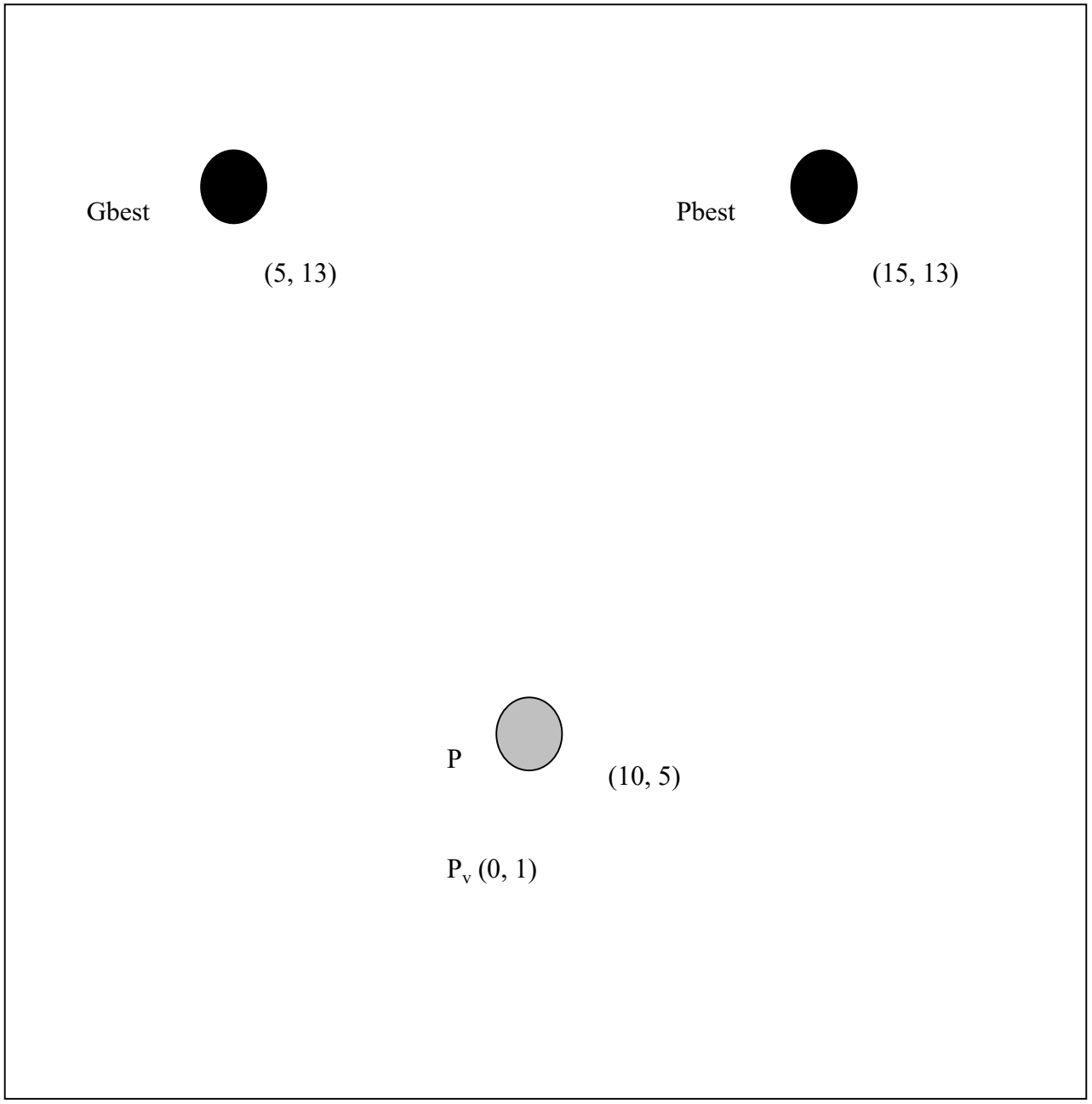
$$\Delta t = 1.0$$

$$V_{id} = \omega V_{id} + C_1 rand(\,)\,(P_{id} - X_{id}) + C_2 Rand(\,)(\,P_{gd} - X_{id})$$

$$c_1 = 1.0$$

$$c_2 = 0.5$$

$$\Delta t = 1.0$$



2. Update velocity for the particle in *x-direction*

$$V_x = V_x + c_1 * rand() * \left(g_{best.x} - P_x\right) + c_2 * rand() * \left(p_{best.x} - P_x\right)$$

$$V_x = 0 + 1.0 * 0.35 * \left(5 - 10\right) + 0.5 * 0.1 * \left(15 - 10\right)$$

$$V_x = 0 + 0.35 * \left(-5\right) + 0.05 * \left(5\right)$$

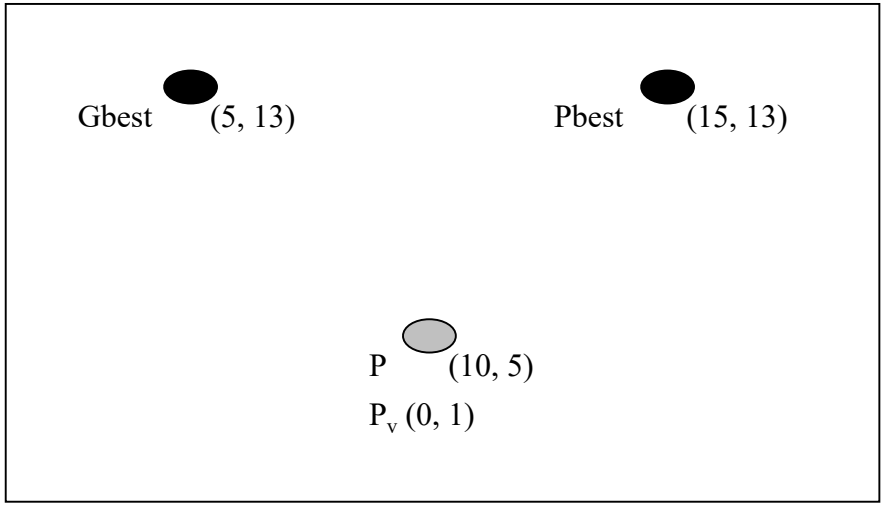
$$V_x = 0 + \left(-1.75\right) + 0.25$$

$$V_x = -1.5$$

$$c_1 = 1.0$$

$$c_2 = 0.5$$

$$\Delta t = 1.0$$



2. Update velocity for the particle in *y-direction*

$$V_y = V_y + c_1 * rand() * \left(g_{best.y} - P_y \right) + c_2 * rand() * \left(p_{best.y} - P_y \right)$$

$$V_y = 1 + 1.0 * 0.2 * \left(13 - 5\right) + 0.5 * 0.45 * \left(13 - 5\right)$$

$$V_y = 1 + 0.2 * \left(8\right) + 0.225 * \left(8\right)$$

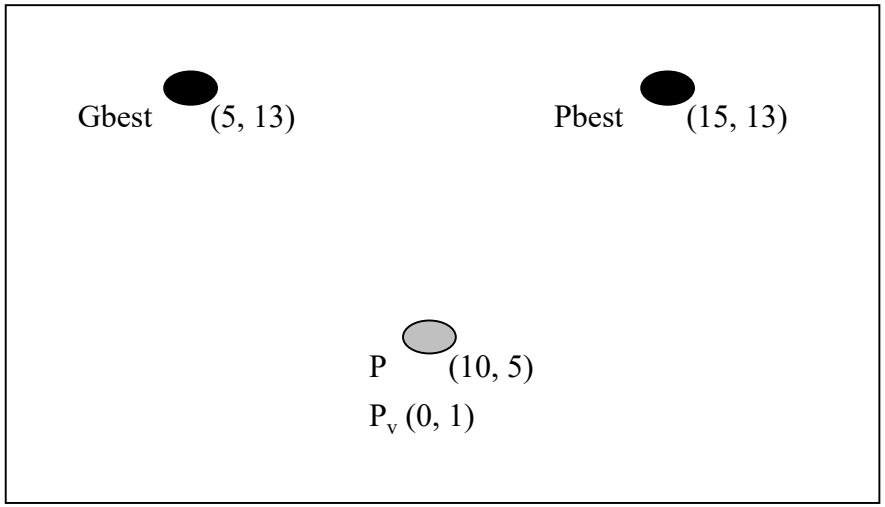
$$V_y = 1 + \left(1.6\right) + 1.8$$

$$V_y = 4.4$$

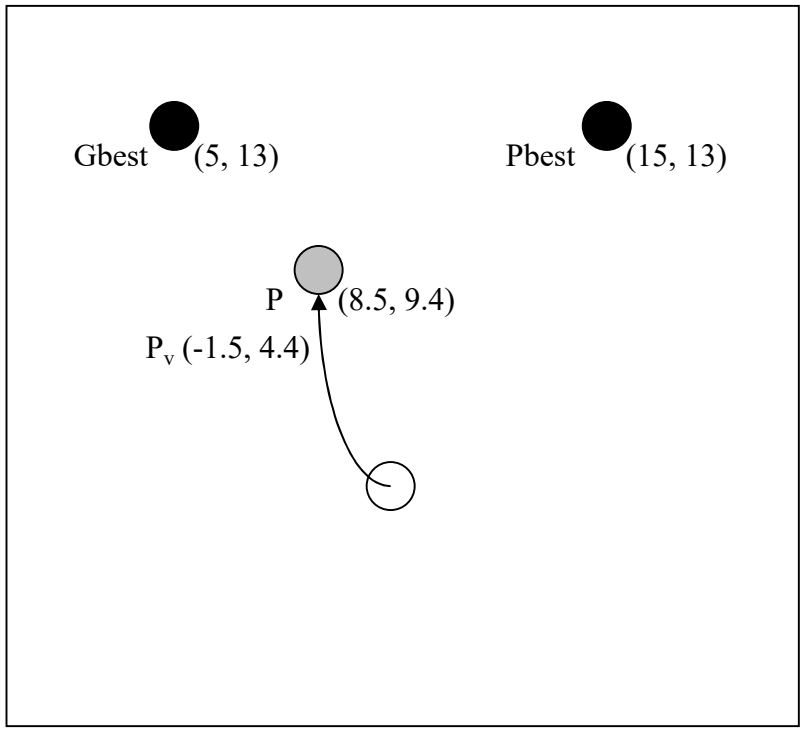
$$c_1 = 1.0$$

$$c_2 = 0.5$$

$$\Delta t = 1.0$$



3. Apply new velocity to update the particle's position



$$P_x = P_x + V_x * \Delta t$$
$$P_x = 10 + (-1.5) * 1.0$$
$$P_x = 8.5$$

$$P_y = P_y + V_y * \Delta t$$
$$P_y = 5 + (4.4) * 1.0$$
$$P_y = 9.4$$

$$X_i^{k+1} = X_i^k + \mathbf{V}_\mathbf{i}^{\mathbf{k+1}} \qquad P = (8.5, 9.4)$$

Updating particle at x-position

4. t = t +1 and update velocities

$$V_x = V_x + c_1 * rand() * \left(g_{best.x} - P_x\right) + c_2 * rand() * \left(p_{best.x} - P_x\right)$$

$$V_x = -1.5 + 1.0 * 0.4 * \left(5 - 8.5\right) + 0.5 * 0.6 * \left(15 - 8.5\right)$$

$$V_x = -1.5 + 0.4 * \left(-3.5\right) + 0.3 * \left(6.5\right)$$

$$V_x = -1.5 + \left(-1.4\right) + 1.95$$

$$V_x = -0.95$$

Updating particle at y-position

$$V_y = V_y + c_1 * rand() * \left(g_{best.y} - P_y \right) + c_2 * rand() * \left(p_{best.y} - P_y \right)$$

$$V_y = 4.4 + 0.1 * 0.2 * \left(13 - 9.4\right) + 0.5 * 0.9 * \left(13 - 9.4\right)$$

$$V_y = 4.4 + 0.02 * \left(3.6\right) + 0.45 * \left(3.6\right)$$

$$V_y = 4.4 + 0.072 + 1.62$$

$$V_y = 6.092$$

Updating new positions (x and y)

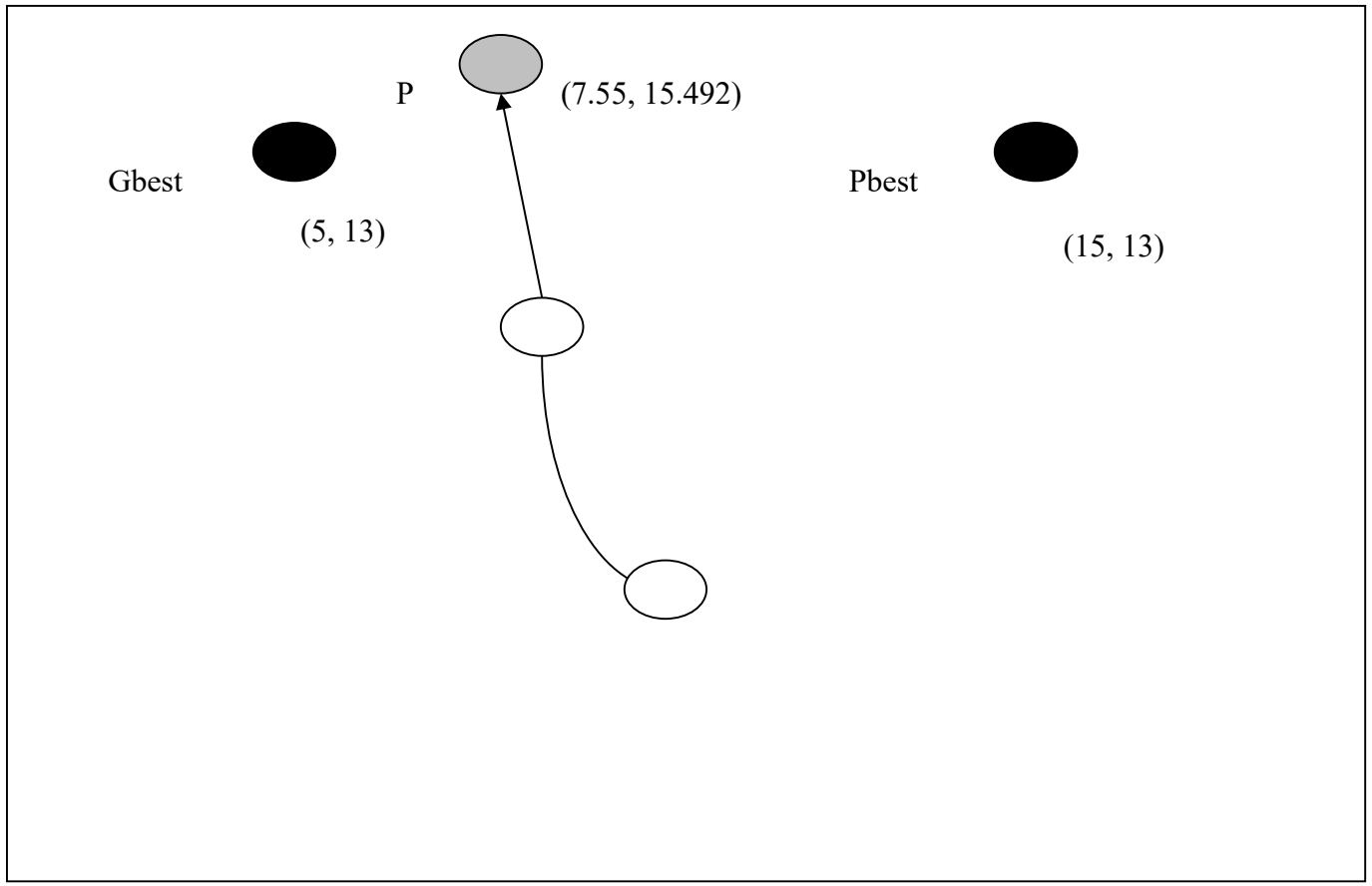
5. Apply new velocities to particle

$$P_x = P_x + V_x * \Delta t$$
$$P_x = 8.5 + (-0.95) * 1.0$$
$$P_x = 7.55$$

$$P_y = P_y + V_y * \Delta t$$
$$P_y = 9.4 + (6.092) * 1.0$$
$$P_y = 15.492$$

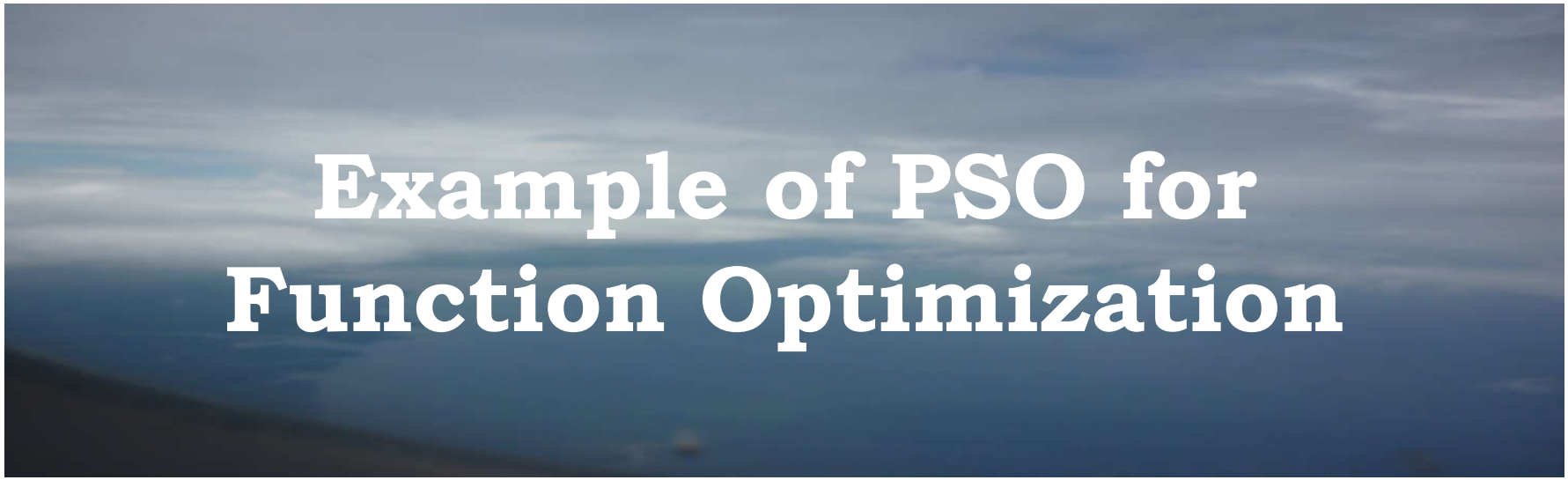
$$P = (7.55, 15.492)$$

Continue updating particle position until it reached to the Gbest...



Fitness calculation

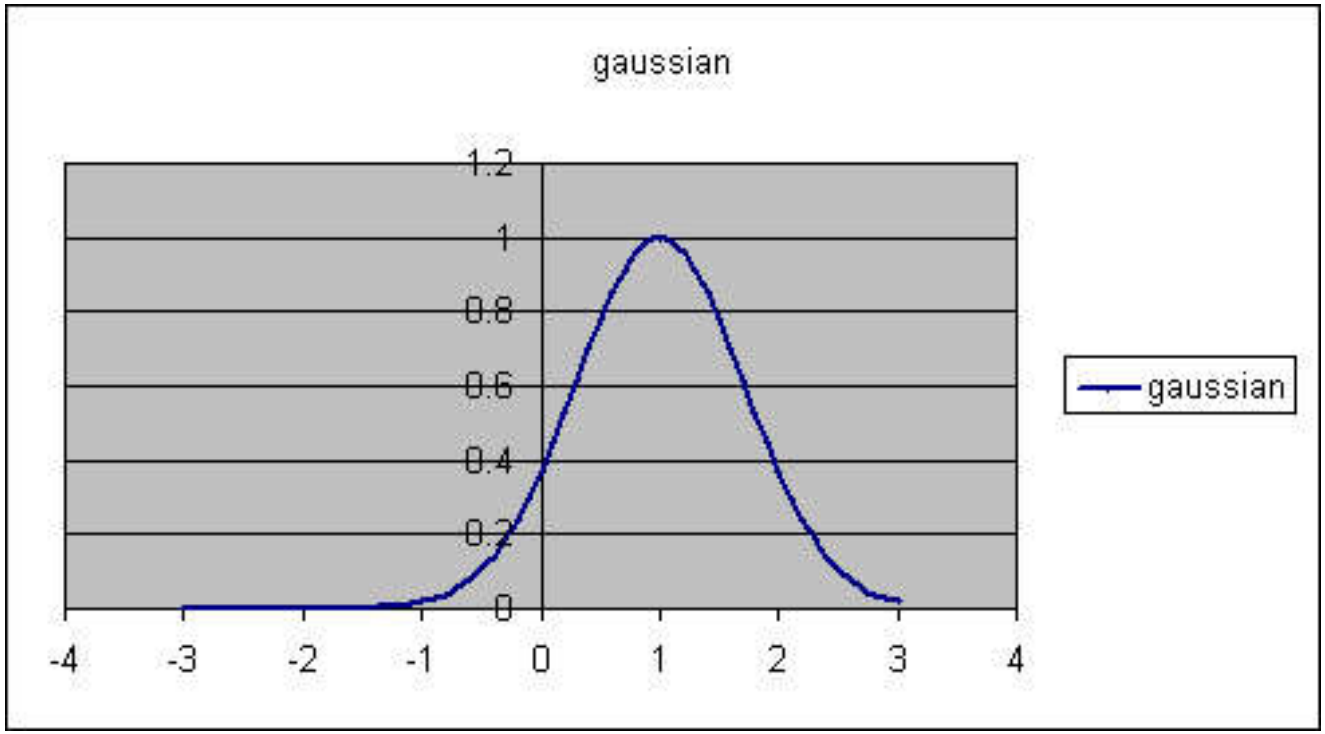
Step 2 of the algorithm (Updating the Velocity) measures the performance of each particle. Here a function is used which measures the closeness of the corresponding solution to the optimum.



Example of PSO for Function Optimization

To optimize the function of $f(x) = e^{-x^2}$

- Given a Gaussian function



Find an optimum value for this function

```
1.6991791 0.6133305
1.7052028 0.6081638
1.7056538 0.6077770
1.6986708 0.6137664
1.6898388 0.6213396
1.6749842 0.6340651
1.6560099 0.6502821
1.6366956 0.6667225
1.6094997 0.6897060
1.5806140 0.7138284
1.5503664 0.7386707
1.5184709 0.7642869
1.4852330 0.7902143
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1.4060761 0.8479804
1.3607036 0.8780014
1.3140050 0.9061059
1.2657387 0.9318187
1.2133817 0.9554893
1.1591577 0.9749870
1.1014002 0.9897707
1.0400160 0.
0.9772844 0.
0.9118850 0.
0.8432645 0.
0.7713039 0.
0.6962639 0.
0.6190003 0.
0.5423361 0.
0.4642272 0.
0.3865573 0.
0.3089193 0.
0.2320093 0.
0.1550930 0.
0.0813752 0.4300424
0.0084367 0.3741129
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-0.1329511 0.2770442
-0.1960644 0.2391725
-0.2542979 0.2073674
-0.3106783 0.1794466
-0.3636364 0.1557498
-0.4107735 0.1366569
-0.4538043 0.1208087
-0.4918866 0.1079890
-0.5242311 0.0979517
-0.5539218 0.0893966
-0.5778323 0.0829469
-0.5980145 0.0777972
-0.6086141 0.0751973
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-0.6244778 0.0714378
-0.6246254 0.0714036
-0.6217737 0.0720677
-0.6137747 0.0739572
-0.5973105 0.0779724
-0.5791411 0.0826049
```

```
1.1608325 0.9744646
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1.1696974 0.9716135
1.1665119 0.9726546
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1.1308252 0.9830304
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1.0935181 0.9912925
1.0703471 0.9950635
1.0419450 0.9982422
1.0086013 0.9999260
0.9746563 0.9993579
0.9374244 0.9960920
0.8964540 0.9893355
0.8514868 0.9781853
0.8035119 0.9621282
0.7550786 0.9417773
0.7046653 0.9164731
0.6546106 0.8875469
0.0371061 0.3956740
0.0071161 0.3731337
-0.0183522 0.3545021
-0.0400986 0.3389830
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-0.0740399 0.3155110
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-0.1211124 0.2845367
-0.1202342 0.2850973
-0.1152943 0.2882632
-0.1094827 0.2920145
-0.1000051 0.2981939
-0.0890754 0.3054146
-0.0764077 0.3139086
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-0.0395916 0.3393407
-0.0191439 0.3539306
0.0020178 0.3693656
0.0278502 0.3886503
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2.7757354 0.0427137
2.7562519 0.0457565
2.7301642 0.0501133
2.6985158 0.0558573
2.6597180 0.0636304
2.6183902 0.0728621
2.5671218 0.0857885
2.5048015 0.1038895
2.4354063 0.1274041
2.3636384 0.1557490
2.2812865 0.1936515
2.1922630 0.2413539
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0.2350328 0.5570085
0.1019293 0.4464040
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-0.1514022 0.2656098
-0.2738318 0.1973755
-0.3926478 0.1437804
-0.5091012 0.1025519
-0.6184246 0.0728540
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-0.8188165 0.0365859
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-0.9884911 0.0191760
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-1.1948707 0.0080873
-1.2440431 0.0065015
-1.2838601 0.0054289
-1.3131071 0.0047459
-1.3326029 0.0043350
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-1.3607733 0.0037981
-1.3582634 0.0038434
```

```
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-1.0298788 0.0162379
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-1.0024546 0.0181366
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-0.8221551 0.0361438
-0.7900890 0.0405825
-0.7545659 0.0460282
-0.7180752 0.0522465
-0.0252058 0.3495716
0.0254667 0.3868512
0.0758924 0.4257194
0.1260098 0.4658652
0.1759472 0.5070919
0.2250292 0.5484936
0.2731247 0.5895783
0.3205914 0.6302769
0.3662959 0.6692612
0.4119481 0.7076505
0.4560870 0.7439054
0.4997967 0.7786424
0.5414856 0.8103934
0.5824394 0.8399971
0.6213212 0.8664095
0.6588550 0.8901370
0.6949682 0.9111530
0.7277587 0.9285646
0.7583256 0.9432664
0.7884852 0.9562475
0.8156147 0.9665735
0.8385360 0.9742663
0.8585589 0.9801932
0.8773916 0.9850796
0.8925352 0.9885177
```

```
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-1.1187407 0.0112312
-1.0827513 0.0130645
-1.0445092 0.0152982
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0.5357299 0.8061006
0.5917963 0.8465125
0.6455619 0.8819443
0.6973701 0.9124839
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0.7986915 0.9602851
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1.1473386 0.9785253
1.1818660 0.9674658
1.2109935 0.9564581
1.2347152 0.9463988
1.2570459 0.9360629
1.2727813 0.9282914
1.2880500 0.9203762
```

```
gbest      Fitness Function (F(x))
0.0000000 1.0000000
```